

DIRECTIONS: THIS EXAMINATION CONTAINS A TOTAL OF **50** MULTIPLE CHOICE QUESTIONS. SELECT CAREFULLY THE BEST ANSWER AND BLACKEN ONLY THE LETTER OF YOUR CHOICE ON THE SEPARATE ANSWER SHEET PROVIDED

TIME ALLOWED: 2:00 HOURS

1. If the tail of vector **B** is placed at the tip of **A** and the tail of **C** is placed at the tip of **B**, then the sum of the three vectors is drawn from the :
A. Tail of A to tip of B C. Tail of B to tip of C
B. Tail of A to tip of C D. Tip of C to tail of B
2. A car is moving towards east in such a way that its position is 20m at time $t = 2\text{sec}$ and 80m at time $t = 6\text{sec}$. what is the average velocity of the car during this time intervals?
A. 15m/sec B. 150m/sec C. 250m/sec D. 25m/sec
3. A bird is flying due east at 10m/s. this statement best describes the birds constant: A. Acceleration B. Direction C. speed D. velocity
4. Two soccer players are approaching each other. Initially, the player has a speed of 8m/s. How much time passes before they collide? A. 4sec B. 6sec C. 3sec D. 8sec
5. A car of mass 1400Kg accelerating from 10m/s to 15m/s over 3.5 sec. What is the average force acting on it? A. 2000N B. 200N C. 20N D. 120N
6. A system is in linear equilibrium if and only if the sum of all:
A. Forces acting on the X axis is zero C. Forces acting on the system is zero
B. Forces acting on the Y axis is zero D. Torque acting on the system is zero
7. A force does a negative work when:
A. The force is opposite to the displacement
B. The force is perpendicular to the displacement
C. Both the force and the displacement are positive
D. Both the force and the displacement are negative
8. A ball is thrown straight up. It rises for a while and then drops back to the ground. Which of the following energy transformations best describe the process?
A. Chemical energy $\rightarrow$ kinetic energy $\rightarrow$ potential energy
B. kinetic energy $\rightarrow$ Chemical energy $\rightarrow$ potential energy
C. potential energy $\rightarrow$ kinetic energy $\rightarrow$ potential energy
D. kinetic energy $\rightarrow$ potential energy $\rightarrow$ kinetic energy
9. What is the purpose of using an inclined plane?
A. Multiplying force C. Multiplying energy
B. Multiplying distance D. Multiplying speed
10. Liquids and gases are collectively called fluids because :
A. Both can flow C. Both have no fixed volume
B. Both can be compressed D. Both fill the volume of their containers
11. The temperature of a gas is most closely relate to :
A. The kinetic energy of translation of its molecule
B. The size of its molecules
C. The potential energy of the molecule

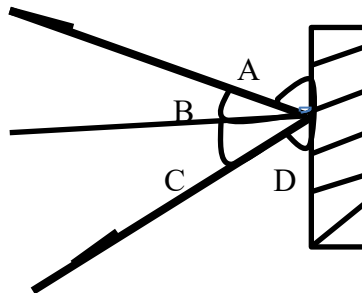
- D. The total energy of its molecules
12. What is the dominant way of heat transfer by which the engine of a car is cooled when the car is in motion?
A. Conduction B. Radiation C. Convection D. Evaporation
13. When a water wave is traveling from deeper to shallower water, what can you observe?
A. The wave speed reduces C. The direction of the wave changes
B. The wave length decreases D. The frequency changes
14. Which quantities decrease as sound travels from water to air?
A. Speed & frequency C. Frequency & wave length
B. Wavelength & speed D. Frequency & period
15. What keeps an earth satellite moving in its orbit?
A. Gravitational force between earth & satellite C. Burring of fuel
B. Ejection of gas from the satellite D. Gravitational attraction of sun
16. Acceleration due to gravity increases if :
A. We go up from the surface of the earth towards its center
B. We go up from the surface of the earth
C. We go from the equator towards the poles
D. The rotational velocity to each increased
17. A wheel of radius 20cm is accelerating at $5\pi \text{ rad/s}^2$. What is the tangential acceleration of a point at the rim of the wheel?
A. $2\pi \text{ rad/s}^2$ B. $1.5\pi \text{ rad/s}^2$ C. π/s^2 D. $0.5\pi \text{ rad/s}^2$
18. If charge Q1 attracts charge Q2 and charge Q2 repels charge Q3. which of the following is correct about their signs?
- | | Q1 | Q2 | Q3 |
|----|----|----|----|
| A. | - | + | - |
| B. | + | - | + |
| C. | - | + | + |
| D. | + | + | - |
19. A 100 watt light bulb is used for 10 hrs. every day. What is the cost of its operation for one month at a fixed rate of 30 cents per **Kwh**?
A. 9 birr B. 3birr C. 15 birr D. 6 birr
20. A circular ring has an area of 100 cm^2 . If the ring is immersed perpendicular in uniform magnetic field of 0.5T. What is the flux through the ring?
A. $5 \times 10^{-4} \text{ Wb}$ B. $5 \times 10^{-2} \text{ Wb}$ C. 50 Wb D. $5 \times 10^{-3} \text{ Wb}$
21. Which law states that the induced **emf** in a closed loop is directly proportional to the rate of the magnetic flux in the loop?
A. Ampere's Law B. Faraday' Law C. Lenz's Law D. Columb's Law
22. Which one of the following is non-magnetic?
A. Zinc B. Steel C. Copper D. Cobalt
23. Digital systems are referred to as binary systems because:
A. They are made from two parts C. They can have only two values

- B. They have two terminals D. Electrons & holes exist in them
 24. For the PnP transistor circuit, the depletion voltage is 0.6v and the base resistance is $280k\Omega$. what is the value of the base current I_B ?

A. $3 \times 10^{-3} A$ B. $3.4 \times 10^{-5} A$ C. $3 \times 10^{-5} A$ D. $3.2 \times 10^{-2} A$

25. Which of the angles shown are compared to the law of reflection?

A. B and C
 B. A and B
 C. B and D
 D. A and C



26. The image of an object formed by a device is always virtual and small. The device may be :

A. Convex lens B. concave lens C. a glass plate D. concave mirror

27. When 999mm is added to 100m, we get:

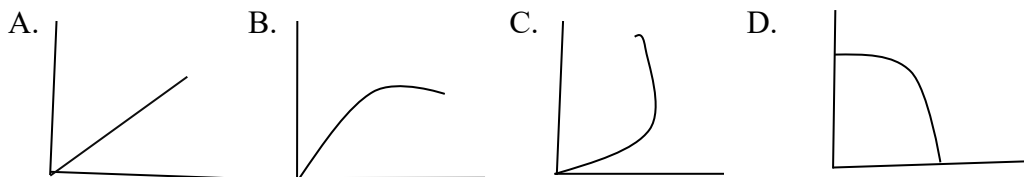
A. 101m B. 100.999m C. 100.99m D. 100.9m

28. Which of the following is not correct?

A. $j \times i = -k$ B. $k \times j = -i$ C. $i \times k = -j$ D. $k \times i = -j$

29. a body travelling with uniform acceleration crosses two points A and B with velocity of 20m/s and 30m/s respectively. The speed of the body at the mid-point of A and B is: A. 24m/s B. 25m/s C. 25.5 m/s D. $10\sqrt{6}m/s$

30. Which of the following graphs represent the distance time variation of a body release from the top of a building?



31. The momentum of the system is conserved:

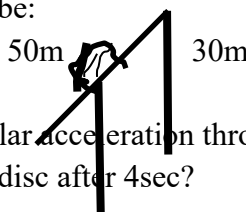
A. Always B. Never C. only in the absence of external force D. Only when an external force acts

32. Which of the following is a form of energy?

A. Pressure B. Momentum C. Light D. Power

33. A block of mass 50 k.g is pulled up along an inclined plane of length 50m. if the coefficient of friction is 0.4, the work done will be:

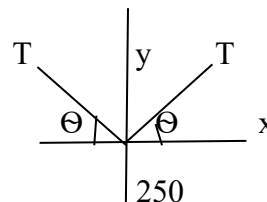
A. 10J B. 23KJ C. 25KJ D. 28KJ



34. A disc initially at rest rotates with constant angular acceleration through an angle of 22.5rad in 3sec. what is the angular speed of the disc after 4sec?

- A. 10rad/s B. 25rad/s C.15 rad/s D. 20 rad/s
35. A body is rotated with a uniform speed in a circle of radius 1m.what is the centripetal acceleration in m/s^2 if it completes 12 rotation in 3sec?
- A. 36π B. $64\pi^2 m/s^2$ C. $72\pi^2$ D. $144\pi^2$
36. The circular motion of a particle with constant speed is: A. Periodic & simple harmonic C. periodic but not simple harmonic B. Neither periodic nor simple harmonic D. simple harmonic but not periodic
37. Traffic lights weighing 250N are suspended from a cable as shown, what is the tension T in the string if $\Theta_1 = \Theta_2 = 30^\circ$. FBD

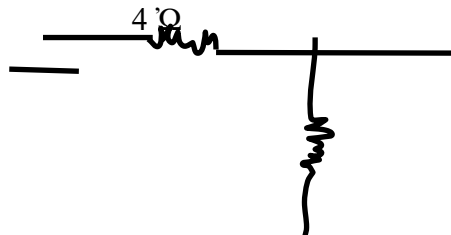
A. 120N B. 165N C. 196N D.250N



38. Bulk modulus of water is $2 \times 10^9 N/m^2$. The change in pressure required to increase the density of water by 0.1% is :
- A. $2 \times 10^9 N/m^2$ B. $2 \times 10^8 N/m^2$ C. $2 \times 10^6 N/m^2$ D. $2 \times 10^4 N/m^2$
39. Heating of water under atmospheric pressure is an: A. Isothermal process B. Adiabatic process C. isobaric process D. Isochoric process
40. If the heat given to a system is fully used in the external work done, then its: A. Internal energy increases B. internal energy decreases C. Temperature remains unchanged D. Temperature increases
41. A particle in SHM has a period of 0.01sec and amplitude of 0.2m. What is the velocity at the center of oscillation? A. 100m/s B. 100π m/s C. 20π m/s D. 40π m/s
42. It is possible to distinguish between transverse and longitudinal waves by studying the property of: A. Interference B. Diffraction C. Reflection D. polarization
43. The sun is visible to us a little before the actual sunrise and a little after the actual sunset. This is because of atmospheric: A. Reflection B. Refraction C. scattering D. Diffraction
44. An electron is released from rest in a region of space with non zero field. As the electron moves, it experience: A. An increasing electric potential C. the same electric potential B. Decreasing electric potential D. Decreasing electric potential energy
45. A capacitor of 20 μ F charged to 500v is connected in parallel with a capacitor of 10 μ F charged to 200v. The common potential is : A.100V B.200V C.300V D.400V

46. A 3 Ω and 6 Ω resistors are connected in parallel. This combination is the n connected with 4 Ω resistor and a 12v battery in series. What is the potential difference a cross the 4 Ω resistors?

A. 4 V B. 8V C.6V D.10V





47. A current carrying power line carries current from west to east. What will be the direction of the magnetic field at a point above the wire? A. East to west B. West to East C. North to South D. South to North
48. If a surface is parallel to magnetic lines of force, then the magnetic flux linked with it is:
A. Small but not zero B. Zero C. Infinite D. Very large
49. An electric motor runs on a DC source of ***emf* 200V** and draws a current of 10A. If the efficiency is 40%, then the resistance of the armature is:
A. 5Ω B. 12Ω C. 120Ω D. 160Ω
50. Dual nature of radiation is shown by:
A. Diffraction and reflection C. Photoelectric effect
B. Refraction and diffraction D. photoelectric effect and diffraction